

# Performance Testing

## Fit Buddy AI – AI Fitness Plan Generator using Gemini Models

Generative AI Project | FastAPI + Google Gemini

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### Test approach

The team tested the local application with Pytest and the FastAPI TestClient, and by running the Uvicorn server to check the main pages, plan generation, feedback update and the nutrition tip endpoint.

### Observed checks

- Application starts and the SQLite database is initialized
- Home page loads and contains FitBuddy
- Health endpoint returns status ok
- Plan generation returns 7 days and a nutrition tip
- Plan update returns an updated 7-day plan
- Nutrition tip endpoint returns a tip
- All 5 automated tests passed in the pytest run (100%)

### Environment

Testing was performed in the local Windows development environment using PowerShell. The tests run without a Gemini API key by using the fallback generator. When Gemini is used, response time depends on the external AI service.